



A Shared Home

How to Live with Friends in **VRCHAT**



by fiaKaiera 

A Vision



Typical VRChat





Kaiera~!

 hang out time?!?!



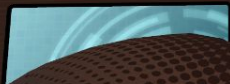
Birb

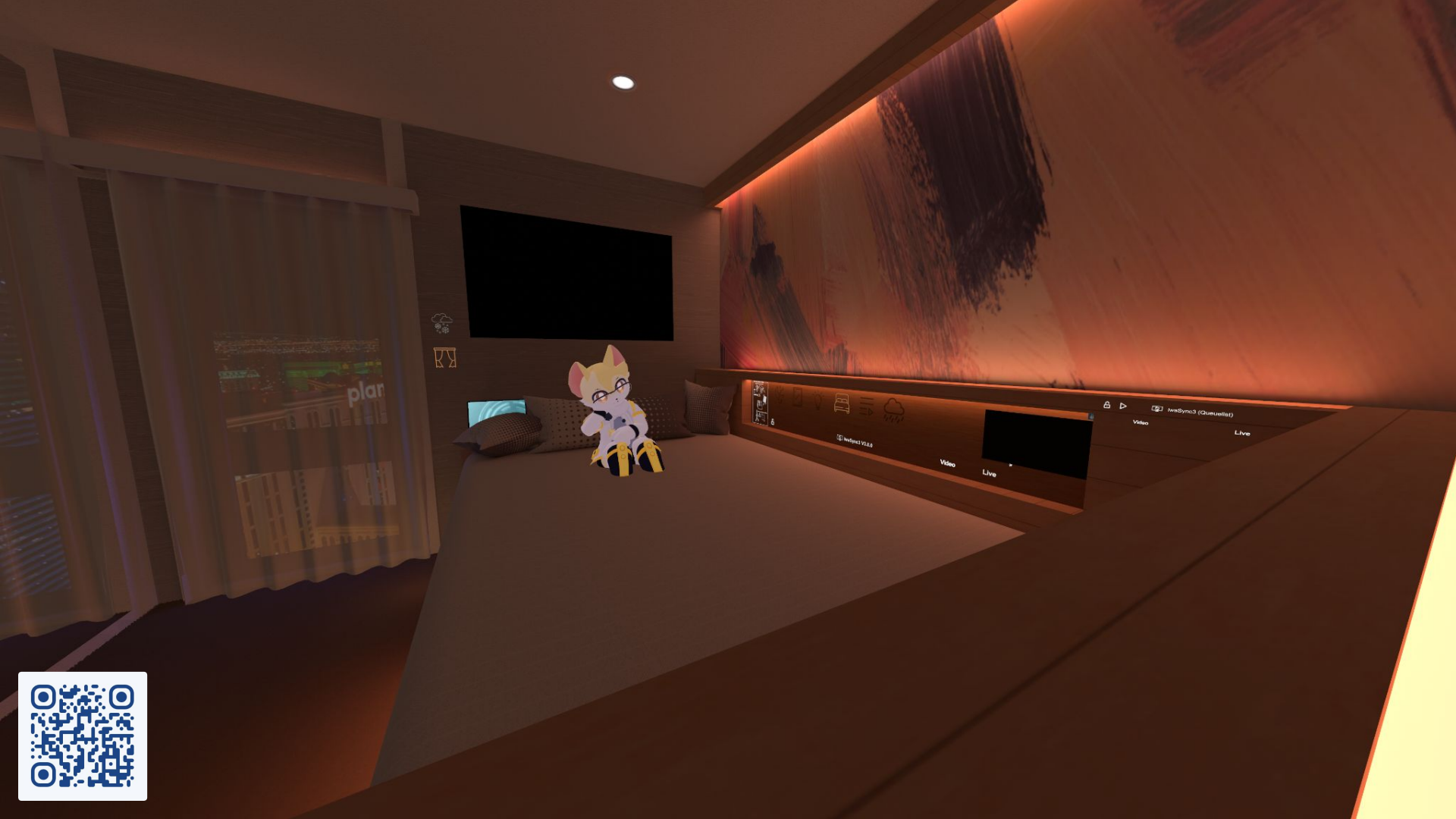
yup yup! gimme a sec O:













birb 9:30 PM

went out to get milk brb



* name is changed, not a serious message



A Shared Home













birb 9:30 PM

went out to get milk brb 🥛

she is still out btw

(her avatar is at her home, came back an hour later)



* name is changed, not a serious message



* picture shared with permission

Meet the Host~!

*Who is **Kaiera**?*



Meet the Host~!

Kaiera @fiaKaiera

- Character Designer
- Multimedia Kemono Artist
- Creator of @ProjectFurria



VTuber at times!

@VKaiera on YT

Art by me!

kaiera.furria.net



In depth VRChat things?!

"Why is an **artist** doing this talk?"



Hosting??

Social Living??

Programming?????

Unity Stuff??



Meet the Host~!

Art by me!

Kaiera @fiaKaiera

kaiera.furria.net

- Character Designer
- Multimedia Kemono Artist
- Creator of @ProjectFurria
- Computer Science Graduate
- Experiments with **Game Development**



Play these at
fiaKaiera.itch.io



VTuber at times!

@VKaiera on YT



A Shared Home

How to Live with Your Friends in **VRCHAT**



by fiaKaiera 

A blurred background image of a VRChat scene, showing various avatars and objects in a virtual environment. The text "by fiaKaiera" is written in a blue, sans-serif font, followed by a small, stylized avatar of a character with orange and yellow fur and a white face.

To make things easy...

This talk is designed to be

SIMPLE

&

ACCESSIBLE

as possible!



← All resources are found **here!**

To make things easy...

Designed to be
[practical]

Concepts will be
[simplified]
(less accurate, more understandable!)

Will have clear
[start points & end points]

Questions? Furity will have a Discord channel for this talk!

And all Q&A answers and common questions will be added in resources~



← All resources are found **here!**

Focus is on two things:

Getting Inside

VRChat-side

Hosting & Sharing With Everyone

The Magic of a Village

World-building

How It Works & Considerations



At the end of the talk...

A quick recap of
everything simplified
from start to finish!



A Shared Home: How to Live with Your Friends in VRChat

Getting Inside

VRChat-side

Hosting & Sharing With Everyone



VRChat-side

Hosting & Sharing With Everyone

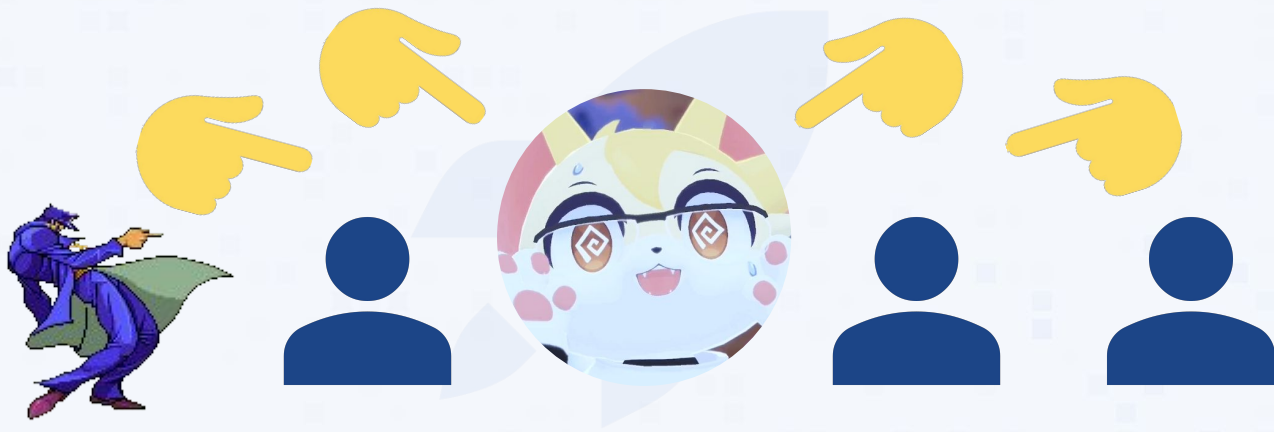
A Shared Home: How to Live with Your Friends in VRChat

Getting Inside



Getting Inside

Part 1: Hosting



Assign a **host**

They will have to do **everything**

...for setup



Getting Inside



What the host does?

- Creating the instance
 - Being the *Instance Owner*
 - Determining the instance type
 - Creating the link to the instance
- "Refreshing" the instance
- Creating a group (recommended but optional)
- Giving everyone access



Getting Inside

Instance Types

Acts slightly differently for living with friends

Invite

Will not work*

**unless the instance is unlocked (risky)*

Friends

Friends only

Also means friends you don't mean to include

Group

Members only

Requires  to create but recommended

Public

Don't.



For Invite+, Friends+, Group+

+ = "...and their friends"



Getting Inside

Instances will "expire"

All instances will end up "out of date"

Caused by:

- VRChat instance updates

Example: "The instance you are trying to join is on an older version" sometimes for non-PC or open beta users

- Specific world author changes

Example: The instance capacity **does not** change even if the world's capacity is updated!! ...*but changes inside the world itself should be fine (geometry, scripts)*



You will have to **close the instance** and re-make them **but not often!**

...or just re-make them without closing but your friends might use the old instance...

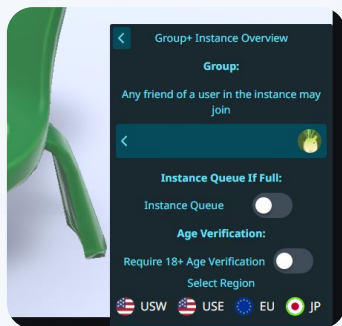


Getting Inside



vrchat.com/home

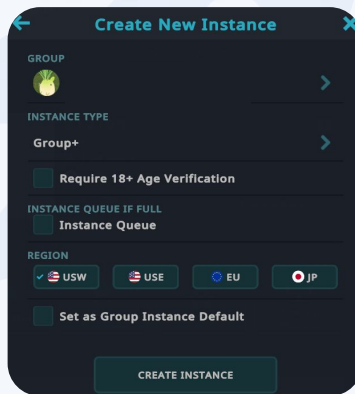
via **VRChat Website**



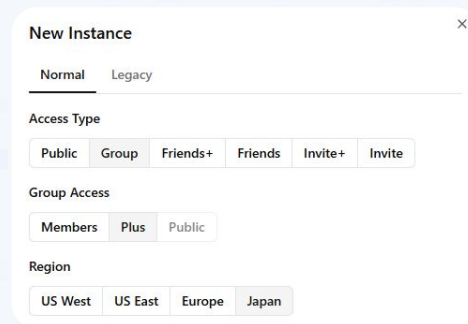
Time to host!

Create the instance!

Inside **VRChat**



A tool like **VRCX!**



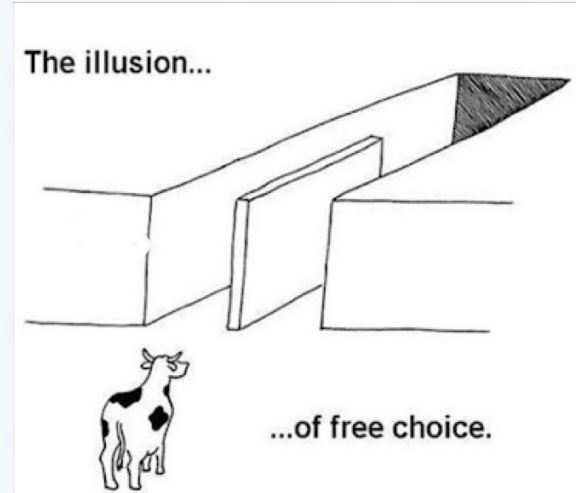
Do this **later** if you have to do it inside VRChat!
We're still in the panel!!!



Getting Inside

But if you want to share...

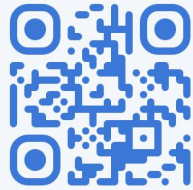
**All roads lead
to the
VRChat website... :(**



Getting Inside

Getting the instance!

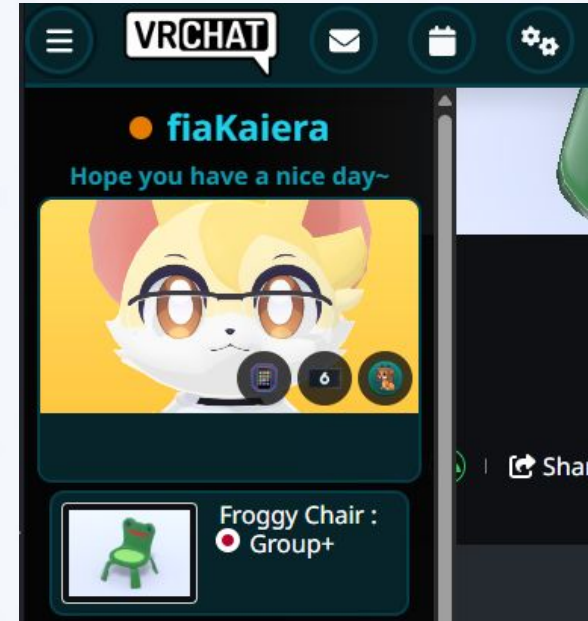
Coming from inside VRChat...



vrchat.com/home

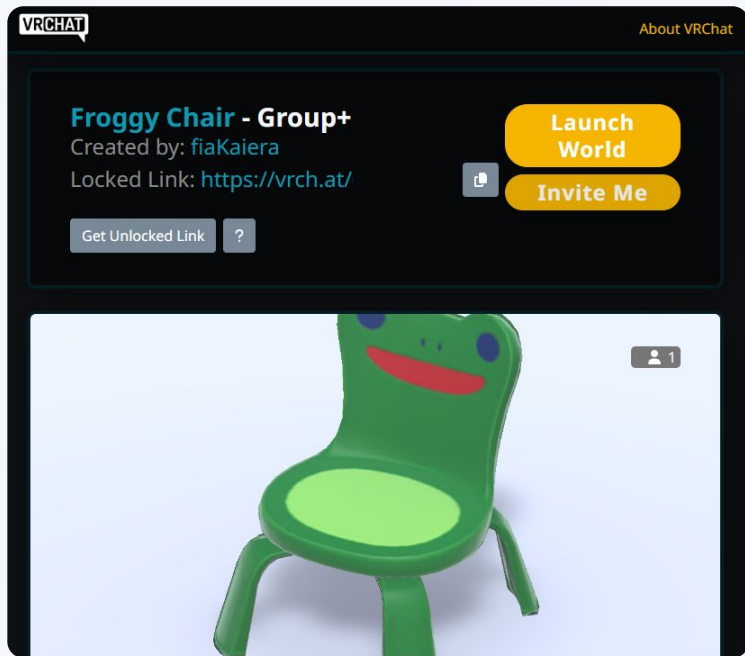
While you are inside the instance...

Click this ->



Getting Inside

You should get something like this...



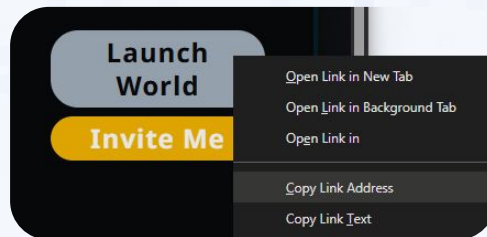
For sharing to non-PC Users:

- Copy the URL or the Locked Link
(URL is better should the locked link expire)

For sharing to PC users:

1. Right-click on "Launch World"
(or hold the button on mobile)
2. Copy the Link Address

The link should start with [**vrchat://**]



Getting Inside

vrchat:// protocol

Source (vrchat.com) World ID Instance ID

`vrchat://launch?ref=vrchat.com&id=wrlD_*****_****_****_*****:#####`

`~group(grp_*****_****_****_*****~groupAccessType(plus)~region(jp)&shortName=*****&attach=1`

Instance Type (Group+ JP Region) Unlocked ID for vrchat.com
(auto-generated, cannot be changed)

Basically...

this is how all the websites and tools launch VRChat!



This is the jargon
that leads to my
friends!

Tip:

You can replace the "#####" (different numbers followed by a "~private", "~friends" or "~group") this with another name that is alphanumeric without spaces (max 32) in the "https://" URL and going there before you copy for a custom name!

You **cannot** use the same name again if you **close** them on the same world & instance type, however.

Only **VRCP** subscribers can ignore that restriction via creating the instance normally. (not using the URL)

Fun Fact:

Steam also uses **steam://** for games!

Getting Inside

The host will give the copied link(s) to everyone!



Part 2: Sharing With Everyone



*"You are **invited** to my abode..."*

"So you don't have to ask me for invites again....."

Unless they block you ig...



Getting Inside

Note:



1. Will **not** be your home world.
(when you "Go Home" in VRChat, it will not be the given world)
2. You are **not** sent to the same instance if it is your home world.
(it will be on your own instance instead, therefore you cannot see them)
3. You will have to **use the link** to re-join,
unless if someone you can see is inside.
(instances are not visible if no one is in them or if fellow instance-mates are not on Online status)
Tip: Use a tool like VRCX if you want to back to previous instances.
4. You cannot join in if the instance is **closed**.
To home with you! >:/

Sidenote:

Me and my friends have a Discord server for private world updates so we do have a shortcut there if it's needed.



Getting Inside

Getting on
PC

...is as easy as making a shortcut.

*Like, **really!***



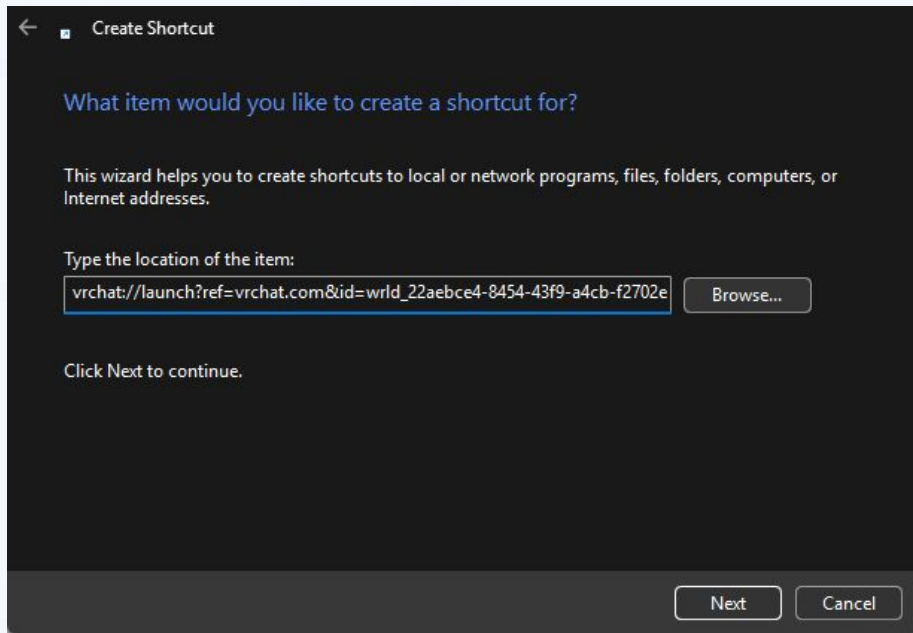
Getting Inside

On Desktop

1. Right-click on Desktop
2. New -> Shortcut
(under "Show More Options" on Windows 11)
3. Paste, Next, Finish
4. ???
5. Done!

Note:

If the link ends with `&attach=1`, it can also show instance details while in VRChat!

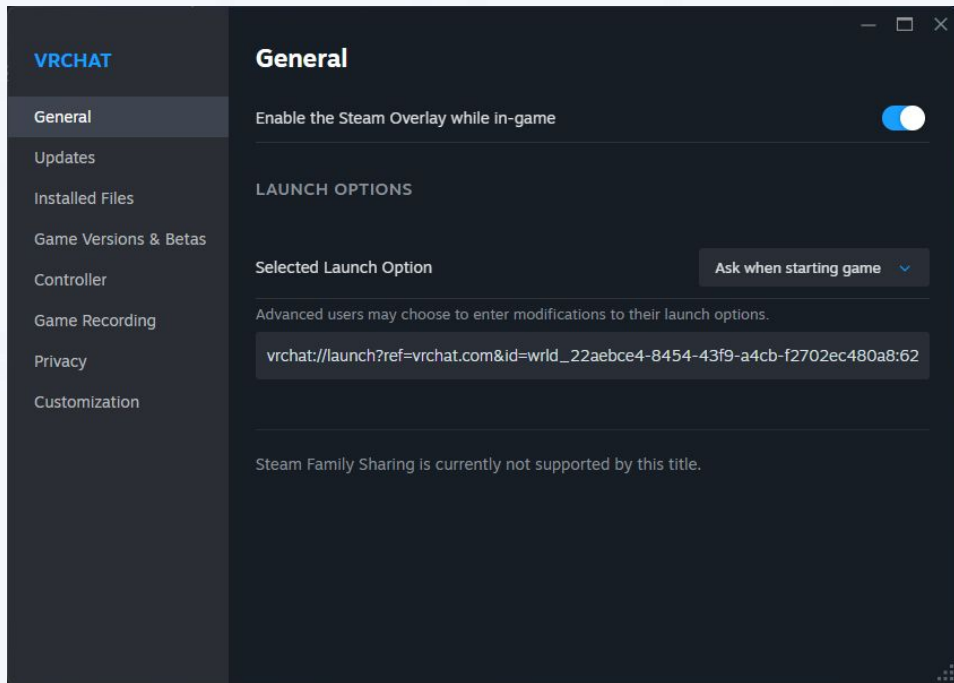


Note for Linux & macOS:

Set this as your launch argument for `Launch.exe` for VRChat instead.



Getting Inside



On Steam

or the Steam Deck / SteamOS

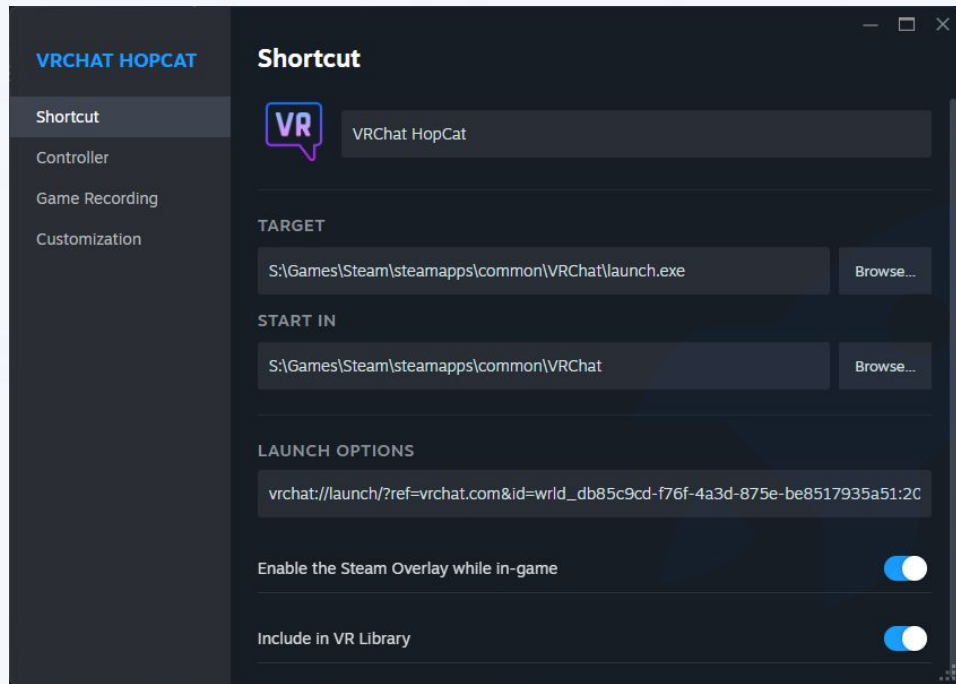
1. Right-click -> Properties
on VRChat in the Library
2. Paste link as Launch Option
3. ???
4. Done!

Note:

This will always boot up VRChat
You will *no longer spawn in your set home*
when starting, unless...



Getting Inside



On Steam (Separated)

or the Steam Deck / SteamOS

1. Right-click -> Manage -> Browse Local Files on VRChat in the Library
2. Note the location of the folder
3. In the Library, add a non-steam Game (found on the very lower left)
4. Select **launch.exe** from VRChat's folder
5. Paste the link as Launch Option (or Target... both do the same)
6. Done!



Turn on **Include in VR Library** to open it from SteamVR!!
Add "**--no-vr**" before the link in Launch Options to disable VR

Getting on **Android / iOS**

...is a bookmark.

A few more steps...



Getting Inside

Why?

VRChat is **not** setup
to run **vrchat://** on mobile.

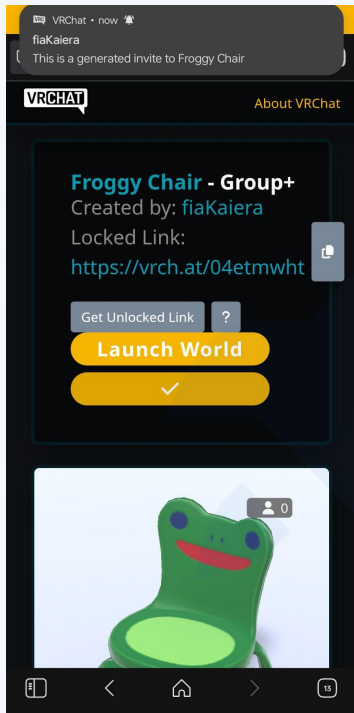
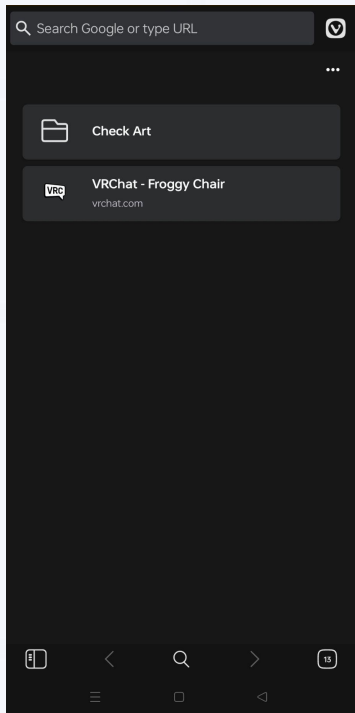
: / indeed...



Getting Inside

On Android / iOS

1. Add the given link as a bookmark and put it on your "Featured Bookmarks" list if any
2. Launch the VRChat app if not opened already
3. Invite Yourself via the bookmark
(it should show as a popup)



Getting Inside

For Android

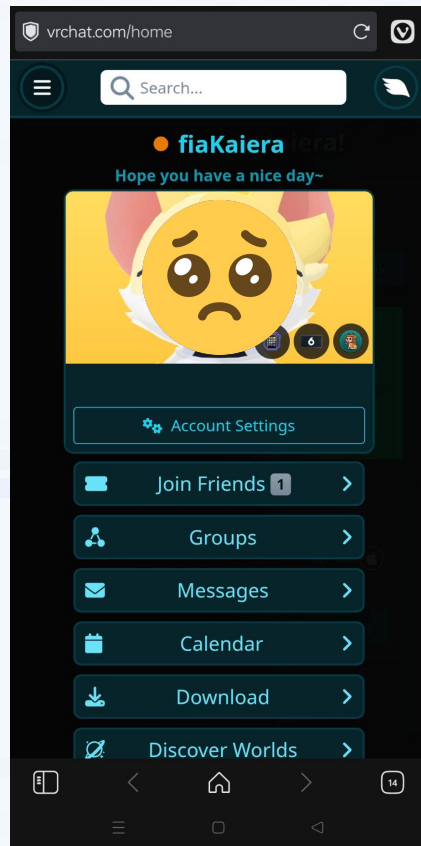
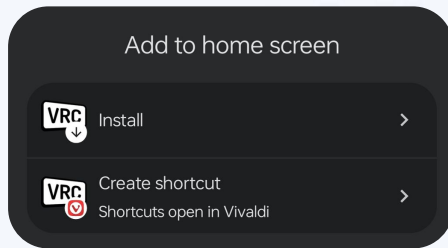
"Add to Home Screen"
is clunky...

It only leads you to vrchat.com/home
instead of the instance.

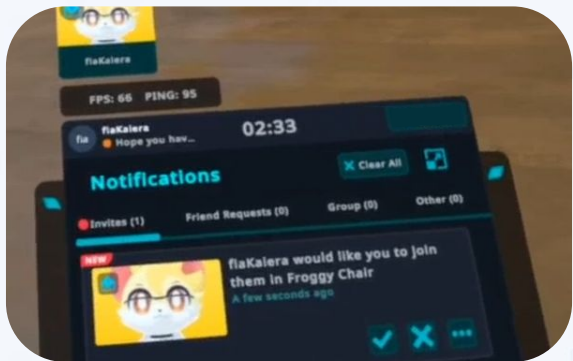
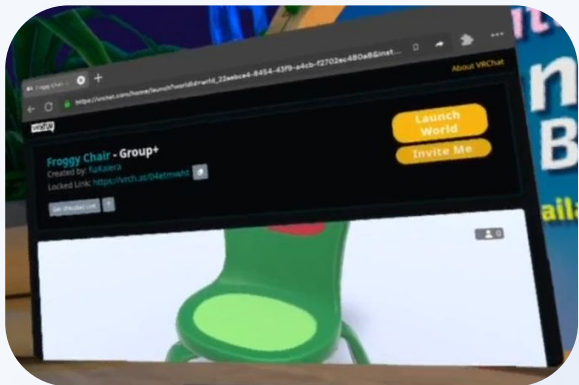
Why?

It treats the browser shortcut as a
"progressive web application", so it strips
all the extras on the link...

You will have to find a "Website Shortcut"
app instead.



Getting Inside



On Quest

You can also use the same methods through the web browser.

Except, you're just inviting yourself within VRChat this time. 🙄

At least you're not waiting for anyone else...

Yes, I do use a Quest 2... via Virtual Desktop.



Getting Inside

What else is this good for?

Pretty much anything that involves
multiple people in one place
constantly...
like **event spaces**!

(Only one-instance events... as it doesn't work on instance queues.
...and if you don't close the instance.)



**Sneak the staff in
before anyone else...**



A Shared Home: How to Live with Your Friends in VRChat

Getting Inside

VRChat-side

Hosting & Sharing With Everyone

- end -



A Shared Home: How to Live with Your Friends in VRChat

The Magic of a Village

World-building

How It Works & Considerations



The Magic of a Village

Part 1: How It Works

Note:

This section will delve a very small bit of **Udon**, just way simplified so you can understand what's happening.



The Magic of a Village

Part 1: How It Works

Unique Spawnpoints!



Unique Spawnpoints

1. Get the player's **display name**
2. If it is them "TeleportTo()" them after a small delay

Ideally after **10 frames**

(1/10th of a second in 60fps)

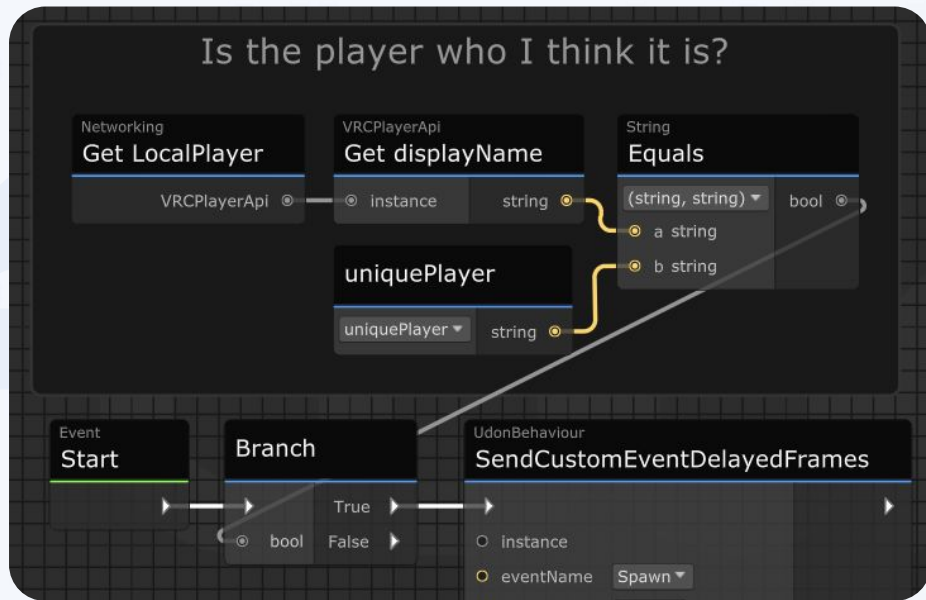
The player will fail to teleport if you do it immediately or too fast.



← Full examples **here!**
(It ended up being too big.)

The Magic of a Village

Udon Graph



Udon#

```
if (localPlayer.displayName == playerName)
    SendCustomEventDelayedFrames(nameof(Spawn), FRAME_DELAY );
// Use localPlayer.TeleportTo() inside Spawn() after
```


The Magic of a Village

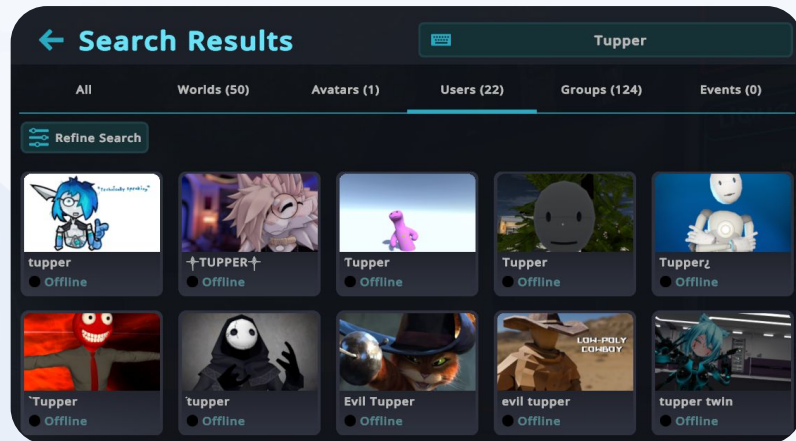
Unique Spawnpoints

Downside

1. Players changing display names
2. Players using the old display name

NOTE:

Udon **cannot** grab user IDs and usernames
(Which I'm pretty sure it's for security purposes)



The Magic of a Village

Solution: PlayerData

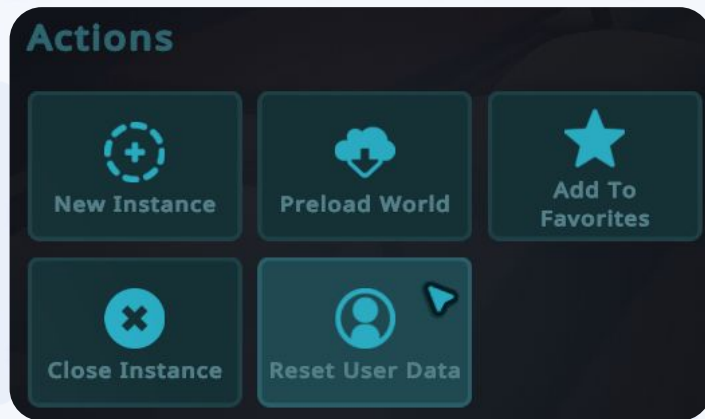
Basically, PlayerData are per-world "cloud save files".

Introduced in November 2025!

NOTE:

Udon **cannot** grab user IDs and usernames
(Which I'm pretty sure it's for security purposes)

Downside:



...or when save data isn't saved
...or if PlayerData gets corrupted or
wiped at some point



← Examples **here!**
(There's multiple ways of doing it!)

A Shared Home: How to Live with Your Friends in VRChat

Now you know how it works!

(Yeah it's all pretty much checking who the player is...)

World-building

How It Works



The Magic of a Village

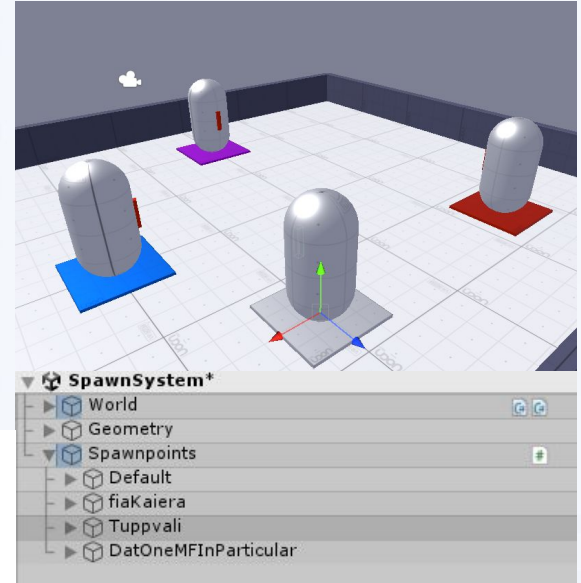
We do have **tools!**



<https://aikatsuyuri.booth.pm/items/4486666>

[Udon Graph] Dynamic Personalized Spawn System

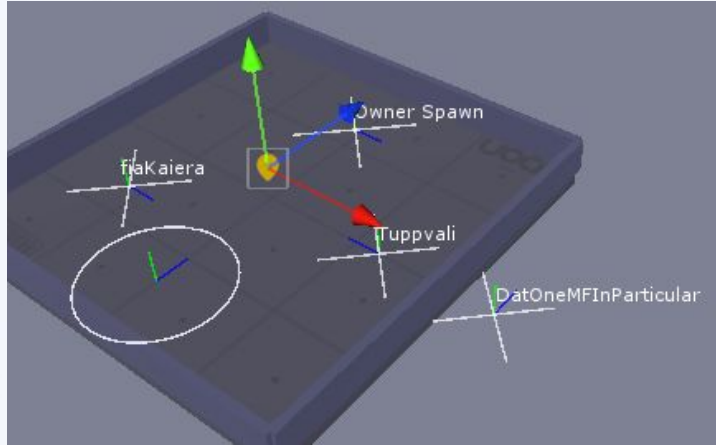
AiTech Dynamic Personalized Spawn System | Version 1.02



← All links are easily accessible **here!**

The Magic of a Village

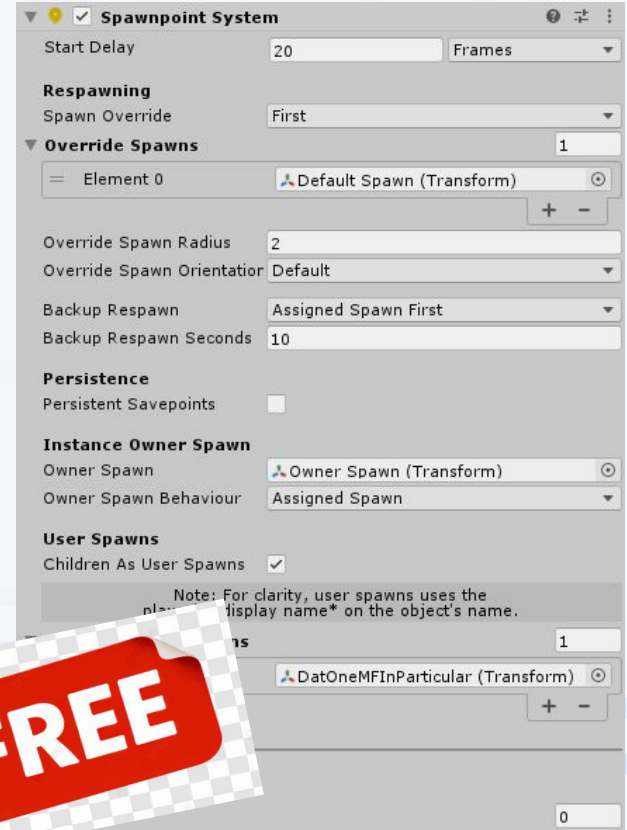
Kaiera's Unique Spawnpoints



Notice:

This is not #AD... It's really just a tool I'm proud of making that I want to share to the public... so maybe it is...

← All links are easily accessible **here!**



The Magic of a Village

Part 2: Considerations



Even so, you're still making a VRChat world.
...and yet with a purpose.



The Magic of a Village

Listen to your residents.

Ask what they want in a home to make it theirs.

...but don't overdo everything. People will understand.

- Have consistent furniture, especially for a small world file size
- Doing everything at once will burn you out
- For the technically inclined and if you're brave enough, you can make a whole custom furniture system thanks to persistence

Technically, you are the HOA...



The Magic of a Village

Privacy is important.

Some like to keep what's theirs safe.

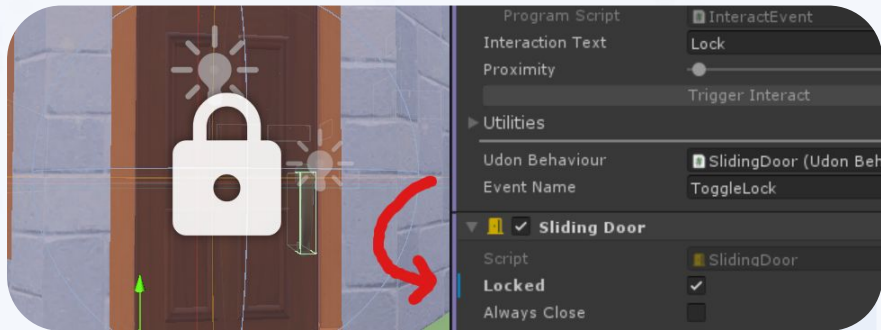
- Access Prevention *(very important for Public)*
- Audio / Voice Zones
- Visibility Zones
- Anti-peeking

BOTTOM TEXT



The Magic of a Village

Access Prevention



If it's not them, don't let them.
Simple solution: Just lock the doors.

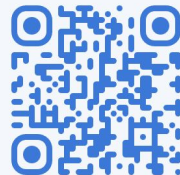
Audio / Voice Zones



You can separate rooms far away or
do Udon scripting magic to isolate voices.



For Voice Zones, there is a powerful tool for this: **Udon Voice Utilities**
<https://github.com/Guribo/UdonVoiceUtils>



The Magic of a Village

Visibility Zones



Literally hide (not cover) the stuff if they are not there or if no one is there.

Anti-peeking



You can use Post-processing Volumes to hide visibility. Alternatively, try to block their view and/or the camera.



The Magic of a Village



Optimization in *very* important.

*Because it's not just avatars lagging you over,
especially for a world made for many.*



The Magic of a Village

VRAM



The Magic of a Village

VRAM



Your PC ran into a problem and needs to restart. We're just collecting some error info, and then we'll restart for you.

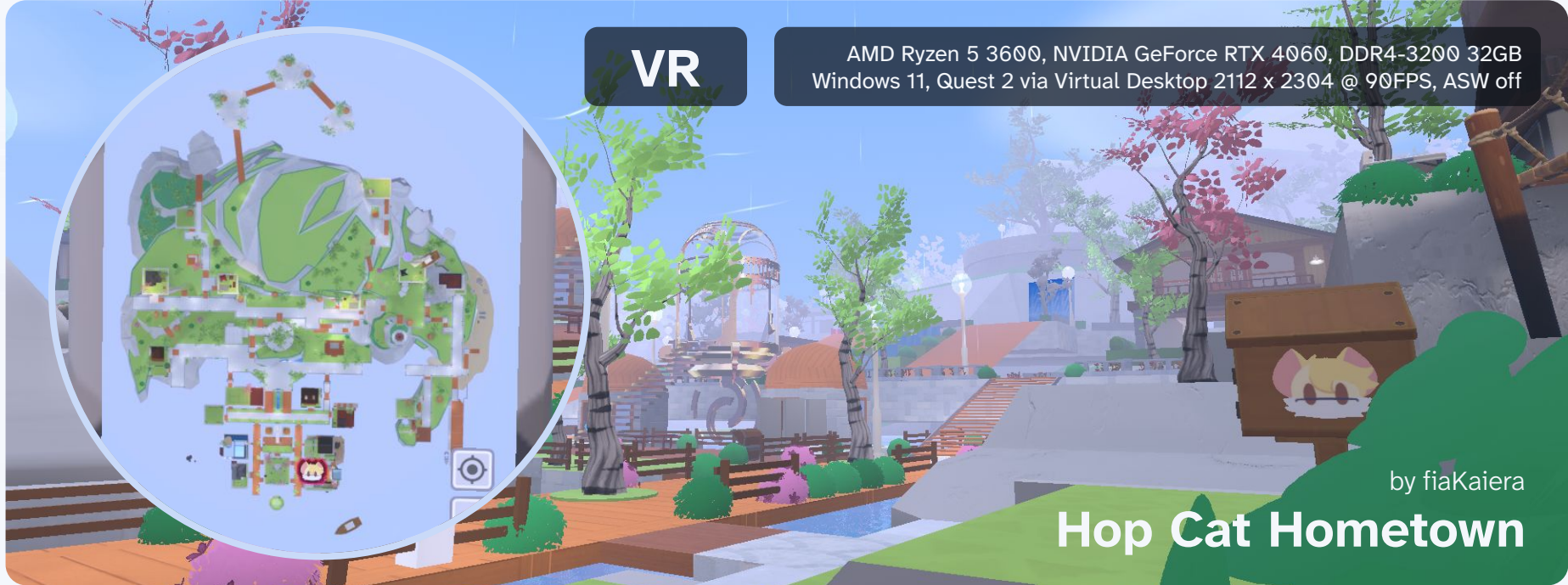
20% complete



For more information about this issue and possible fixes, visit <https://www.windows.com/faq>.
Microsoft Windows | Windows 10 | Windows 11 | Windows 12



The Magic of a Village

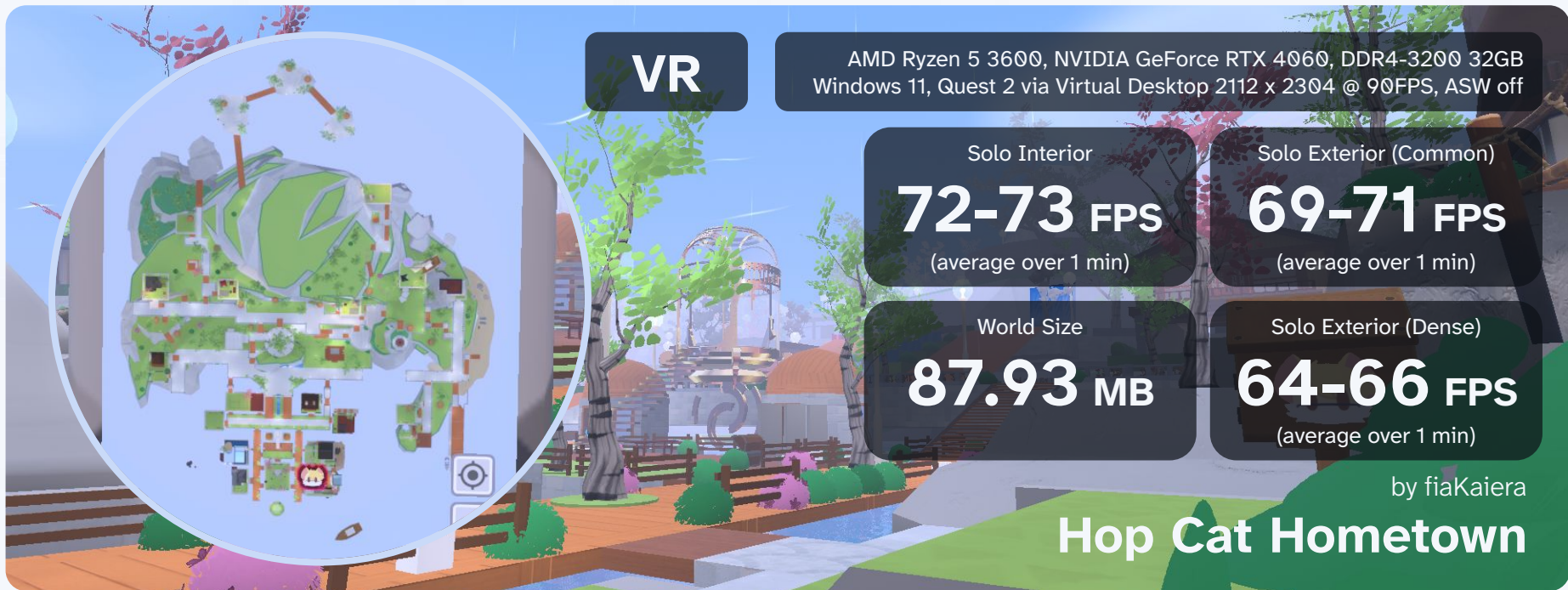


*info as of May 10, as Hop Cat Hometown is regularly updated, calculations are done through observation from the VRChat full settings debug section in-game



← Public version world link **here** if you want to visit!
(PC Only)

The Magic of a Village



VR

AMD Ryzen 5 3600, NVIDIA GeForce RTX 4060, DDR4-3200 32GB
Windows 11, Quest 2 via Virtual Desktop 2112 x 2304 @ 90FPS, ASW off

Metric	Value
Solo Interior	72-73 FPS (average over 1 min)
Solo Exterior (Common)	69-71 FPS (average over 1 min)
World Size	87.93 MB
Solo Exterior (Dense)	64-66 FPS (average over 1 min)

by fiaKaiera

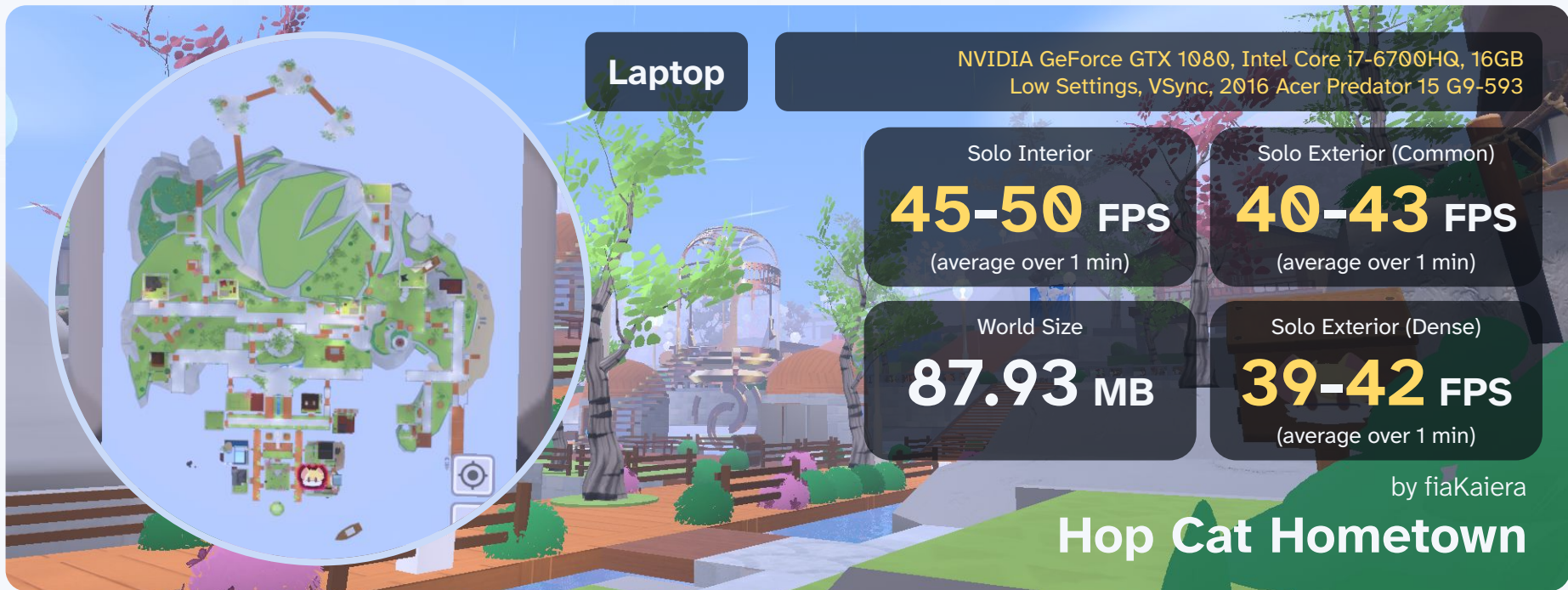
Hop Cat Hometown

*info as of May 10, as Hop Cat Hometown is regularly updated, calculations are done through observation from the VRChat full settings debug section in-game



← Public version world link **here** if you want to visit!
(PC Only, 71.11 MB)

The Magic of a Village



*info as of May 10, as Hop Cat Hometown is regularly updated, calculations are done through observation from the VRChat full settings debug section in-game



← Public version world link **here** if you want to visit!
(PC Only, 71.11 MB as of May 10)

The Magic of a Village

Optimization is **very** important.

Try to achieve as much FPS when you can.

- Try to aim for **Max FPS** for **solo in VR**
 - If not, at least **60 FPS** or **75 FPS**
 - For Desktop only, note VR is rendered 3x than normal (screen, left eye and right eye)
- As soon as people come in
The FPS can **quickly go down** depending on an avatar's performance...
Example: 60 FPS in VR -> **29 FPS** with 11 people on a friend's setup

Try and not follow this:

Performance Tips

- Aim for at least 45 FPS with a single VR user at the spawn.

From: VRChat Creation Docs



IMPORTANT:

LOW FPS IN VR CAN AND WILL CAUSE **NAUSEA**



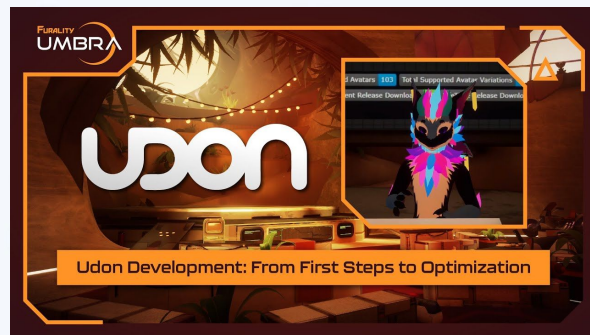
The Magic of a Village

Optimization is **very** important.

...but it's not the main topic of this talk.



by MyroP



by Happyrobot33

...and more!



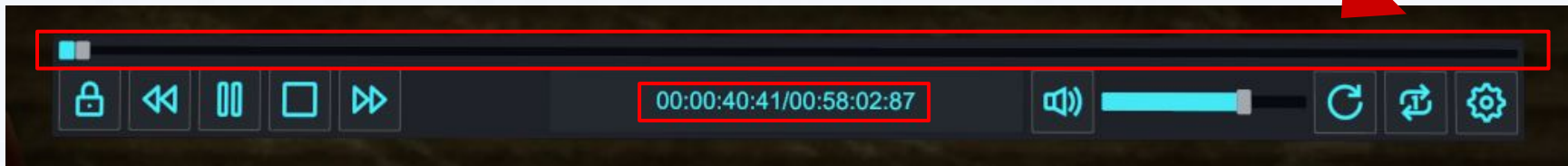
← All easily accessible **here!**

FF14 low-poly grapes my beloved...



The Magic of a Village

Be careful of placing video players...



These update **every frame for each one** even when **disabled!** (Many video players do this!!)
Keep one*, **teleport** them, and do **NOT** place multiple copies!

*Multiple screens are fine, except when they hide UI (YamaPlayer, VizVid, etc.)
It's only parts that update video progress visually affect performance



← All easily accessible **here!**

A Shared Home: How to Live with Your Friends in VRChat

Summary



Summary

Getting Inside

VRChat-side

- Copy a link from vrchat.com/home
- Make a shortcut on your PC

The Magic of a Village

World-building

- Use Unique Spawnpoints
- Do not forget the ice cream
- Respect Privacy
- Low FPS can make people sick



← More info as always **here!**

*Sometimes, just having a place to call **home**
can sometimes have an impact on how we interact
with each other.*

To hang out with, and not just a party every weekend.

*Let's make **#SharedSpace** worlds for those who can't make homes but
want for creatures who live far away to be even closer~*



Thanks for coming~!



Big shout out to
Wishdream
For helping with this!

Check out their new avatar!

Kinu (shown on the left)

Available at
mirror.wishdream.org



← All resources are found **here!**



kaiera.furria.net

@fiaKaiera

